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END SEMESTER EXAMINATION 2025

Name of the Program: B.Tech Petroleum Engineering

Semester: VI

Name of the Course: Alternate Energy Resources

Course Code: TPE-611

Time: Three Hours

MM: 100

Note:

- All questions are compulsory.
- Answer any two sub questions in each main question.
- Total marks for each main question is **twenty**
- Each Question carries **10 marks**

Q1		COs
a.	Discuss in details about the non-conventional energy sources and their advantages and estimated potential?	CO1, CO2
b.	What are the renewable energy resources? Discuss advantages and prospects of renewable energy resources	CO1, CO2
c.	Discuss the solar radiation geometry specifying the different angle used in energy calculations with the help of diagrams?	CO3
Q2		
a.	Discuss the classification solar energy storage systems? Hence explain the basic equation for the energy storage unit?	CO3, CO4
b.	Discuss in details about the types of collectors? Explain the performance analysis of flat plate collectors (FPC)	CO1, CO4
c.	Illustrate the following storage a. Battery storage b. Capacitor storage c. Inductor storage	CO1, CO2
Q3		
a.	Discuss high head power plants in hydroelectric power plants with labeled diagram?	CO1
b.	How one can select the site for hydroelectric power plants? Discuss	CO1, CO2
c.	Discuss the classifications of hydraulic turbines? Discuss the turbines on the basis of specific speed?	CO1, CO5
Q4		
a.	What do you mean by wind and wind energy? Discuss its characteristics advantage and disadvantages	CO1, CO5
b.	Explain sources or origins of wind? Explain principle of wind energy conversion and wind power?	CO4, CO5
c.	What is wind energy pattern factor? Discuss the basic components of wind energy conversion system?	CO3, CO6
Q5		

a.	Discuss factors affecting generation of biogas? Explain the main components of a biogas plant	CO4, CO6
b.	With the help of neat and labeled diagram discuss the floating drum type biogas plant?	CO4, CO6
c.	Tabulate the conversion between fixed dome type and movable drum type biogas plants on the basis of different aspects?	CO2, CO3